

P3: Writing your Application

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Relational Schema

Artifacts(artifact_id, name, description, date_of_creation)

Paintings(artifact_id, medium, paint_type) foreign key artifact_id references Artifacts

Photographs(artifact_id, film_type) foreign key artifact_id references Artifacts

Sculpture(artifact_id, material, dimensions) foreign key artifact_id references Artifacts

Exhibitions(exhibition_id, name, description, start_date, end_date)

Displays(artifact_id, exhibition_id, room) foreign key artifact_id references Artifacts and
foreign key exhibition_id references Exhibitions

Artists(artist_id, name, birth)

Creates(artifact_id, artist_id) foreign key artifact_id references Artifacts and foreign key
artist_id references Artists

Genres(genre_name)

Exhibition_Genre(exhibition_id, genre_name) foreign key exhibition_id references

Exhibitions and foreign key genre_name references Genres

Artifact_Genre(artifact_id, genre_name) foreign key artifact_id references Artifacts and
foreign key genre_name references Genres

Tours(tour_id, description, cost, employee_id) foreign key employee_id references

Museum_Personnel

Tour_Exhibitions(exhibition_id, tour_id) foreign key exhibition_id references Exhibition and
foreign key tour_id references Tours

Museum_Personnel(employee_id, name)

Organizer(employee_id, tour_id) foreign key employee_id references Museum_Personnel
and foreign key tour_id references Tours

TourDates(date, start_time, tour_id, end_time) foreign key tour_id references Tours

Visitor(email, name, address, birth)

Reservation(email, booking_date, tour_id, date, start_time) foreign key email references
Visitor, foreign keys date, start_time references TourDates, tour_id references Tours

Stored procedure

Description

The stored procedure is used to organize new exhibitions efficiently. It takes a target exhibition and a genre name as inputs. The procedure identifies all artifacts belonging to that specific department that are currently in storage. For each artifact, it looks at the date of creation using a cursor. By looping through the artifacts and using conditional logic, the artifacts are assigned rooms based on their year of creation:

- 1) Ancient Wing: Art created before 1500.
- 2) Classical Hall: Art between 1500 and 1899
- 3) Modern Gallery: Art from 1900 onwards
- 4) Other Arts: Art with unknown dates of creation,

The procedure would then perform inserts into the Displays table and return the number of artifacts added to the new exhibition.

Code

```
CREATE PROCEDURE ORGANIZE_EXHIBITION (IN target_exhibition_id INTEGER,
IN dept_name VARCHAR(50), OUT total_moved INTEGER)
LANGUAGE SQL
BEGIN
    DECLARE current_id INTEGER;
    DECLARE year_made INTEGER;
    DECLARE assigned_location VARCHAR(30);
    DECLARE at_end INT DEFAULT 0;
    DECLARE move_count INTEGER DEFAULT 0;
    DECLARE not_found CONDITION FOR SQLSTATE '02000';

    -- cursor to find artifacts in storage based on genre/department
    DECLARE C1 CURSOR FOR
        SELECT a.artifact_id, a.date_of_creation
        FROM Artifacts a
        JOIN Artifact_Genre ag ON a.artifact_id = ag.artifact_id
        WHERE ag.genre_name = dept_name
        AND a.artifact_id NOT IN (SELECT artifact_id FROM Displays);

    DECLARE CONTINUE HANDLER FOR not_found SET at_end = 1;

    OPEN C1;

    FETCH C1 INTO current_id, year_made;

    WHILE at_end = 0 DO
        -- chronological room assignment
```

```

    IF (year_made IS NULL) THEN
        SET assigned_location = 'Research Lab';
    ELSEIF (year_made < 1500) THEN
        SET assigned_location = 'Ancient Wing';
    ELSEIF (year_made >= 1500 AND year_made < 1900) THEN
        SET assigned_location = 'Classical Hall';
    ELSE
        SET assigned_location = 'Modern Gallery';
    END IF;

    -- insert in the displays table
    INSERT INTO Displays (artifact_id, exhibition_id, room)
    VALUES (current_id, target_exhibition_id, assigned_location);

    SET move_count = move_count + 1;

    FETCH C1 INTO current_id, year_made;
END WHILE;

CLOSE C1;
SET total_moved = move_count;
END @

```

Screenshot running procedure

The question mark is used as a placeholder for the out parameter total_moved. This allows db2 to return the value of this parameter, indicating how many artifacts were processed and moved to the new exhibition.

```

cs421g22@winter2026-comp421:~$ db2 "CALL ORGANIZE_EXHIBITION(12, 'Renaissance', ?)"

Value of output parameters
-----
Parameter Name  : TOTAL_MOVED
Parameter Value : 4

Return Status = 0

```

Screenshot of effect

To demonstrate the procedure's logic, we initialized a new exhibition (ID 12) in the Exhibitions table. We intentionally introduced a test set of artifacts under the 'Renaissance' genre, including items with pre-1500 and post-1900 creation dates. This allows us to showcase the procedure's ability to perform an automated chronological sort, distributing items into the 'Ancient Wing,' 'Classical Hall,' and 'Modern Gallery' based on the date of creation.

```

cs421g22@winter2026-comp421:~$ db2 "SELECT * FROM Displays WHERE exhibition_id = 12"

ARTIFACT_ID EXHIBITION_ID ROOM
-----
0 record(s) selected.

cs421g22@winter2026-comp421:~$ db2 "CALL ORGANIZE_EXHIBITION(12, 'Renaissance', ?)"

Value of output parameters
-----
Parameter Name : TOTAL_MOVED
Parameter Value : 4

Return Status = 0
cs421g22@winter2026-comp421:~$ db2 "SELECT * FROM Displays WHERE exhibition_id = 12"

ARTIFACT_ID EXHIBITION_ID ROOM
-----
        6          12 Ancient Wing
       14          12 Classical Hall
       49          12 Classical Hall
       81          12 Modern Gallery

4 record(s) selected.

```

Application program

Menu:

```

cs421g22@winter2026-comp421:~/java$ java Museum
Group 22 Museum Main Menu:
    1. Search for an artifact
    2. Make a reservation for a tour
    3. Relocate an artifact
    4. View tour statistics
    5. Add a new exhibition
    6. Quit
Select an option: |

```

Option 1: Searching for artifacts by artist, genre or time period

```

[Select an option: 1
    a) Search by Artist
    b) Search by Genre
    c) Search by Time Period
Select a search option by typing a, b, or c: |

```

Sub-option a)

```
[Select a search option by typing a, b, or c: a
[You selected to search by artist. Insert the name of the artist: Vinci
11 | The Mona Lisa | Leonardo da Vinci
13 | The Virgin and Child with Saint Anne | Leonardo da Vinci
14 | Saint John the Baptist | Leonardo da Vinci
15 | La Belle Ferronniere | Leonardo da Vinci
40 | Portrait of Isabella d'Este | Leonardo da Vinci
```

Sub-option b)

```
[Select a search option by typing a, b, or c: b
[You selected to search by genre. Insert the genre: Baroque
8 | Fame Riding Pegasus | Baroque
9 | The Marly Horses | Baroque
10 | Ceres | Baroque
17 | The Lacemaker | Baroque
21 | The Astronomer | Baroque
24 | Death of the Virgin | Baroque
31 | Portrait de Philippe IV, roi d'Espagne, en pied | Baroque
32 | Portrait of a Seated Woman | Baroque
41 | Bathsheba at Her Bath | Baroque
42 | The Slaughtered Ox | Baroque
43 | L'Archange Raphael quittant la famille de Tobie ou Tobie et sa famille prosternes devant lange | Baroque
46 | A Lesson in Humility | Baroque
48 | The Infant Jesus Playing with a Nail | Baroque
59 | e Combat du Tessin, de la tenture de l'Histoire de Scipion | Baroque
92 | Portrait d'homme | Baroque
```

Sub-option c)

```
[Select a search option by typing a, b, or c: c
[You selected to search by time period. Insert the start year for this time period:
[1600
Insert the end year for this time period:
[1700
6 | Sleeping Hermaphroditus | 1620
17 | The Lacemaker | 1670
21 | The Astronomer | 1668
24 | Death of the Virgin | 1606
31 | Portrait de Philippe IV, roi d'Espagne, en pied | 1624
32 | Portrait of a Seated Woman | 1653
41 | Bathsheba at Her Bath | 1654
42 | The Slaughtered Ox | 1655
43 | L'Archange Raphael quittant la famille de Tobie ou Tobie et sa famille prosternes devant lange | 1637
46 | A Lesson in Humility | 1671
48 | The Infant Jesus Playing with a Nail | 1665
59 | e Combat du Tessin, de la tenture de l'Histoire de Scipion | 1688
67 | Casque orne d'une tete de loup | 1685
69 | Casque orne d'une tete de dauphin | 1685
70 | Bouclier retourne | 1685
71 | Pertuisane | 1685
72 | Bouclier aux lions | 1685
73 | Casque orne d'un masque de feuillage | 1685
74 | Bouclier orne d'un foudre | 1685
75 | Hallebarde | 1685
91 | Portrait de Marie Amelie Josephine, archiduchesse d'Autriche, infante, duchesse de Parme | 1700
92 | Portrait d'homme | 1658
```

Option 2: Making a reservation (submenu)

This option has two parts. 1) The available tours are queried and a submenu is generated with those options. 2) The user selects the tour, inputs their email and the reservation is added to the database with another query.

Select an option: 2

[You have selected to make a reservation.

These are the available tours:

1		exhibition02		2026-04-10		10:00:00-11:00:00		\$7.5
2		exhibition02		2026-04-10		12:00:00-13:00:00		\$7.5
3		exhibition02		2026-04-10		14:00:00-15:00:00		\$7.5
4		Ancient civilizations		2026-04-15		10:00:00-11:30:00		\$12.0
5		Renaissance masters		2026-04-15		10:00:00-11:30:00		\$12.0
6		exhibition02		2026-04-15		10:00:00-11:00:00		\$7.5
7		Ancient civilizations		2026-04-15		12:00:00-13:30:00		\$12.0
8		Renaissance masters		2026-04-15		12:00:00-13:30:00		\$12.0
9		exhibition02		2026-04-15		12:00:00-13:00:00		\$7.5
10		Ancient civilizations		2026-04-15		14:00:00-15:30:00		\$12.0
11		Renaissance masters		2026-04-15		14:00:00-15:30:00		\$12.0
12		exhibition02		2026-04-15		14:00:00-15:00:00		\$7.5
13		Ancient civilizations		2026-04-20		10:00:00-11:30:00		\$12.0
14		Renaissance masters		2026-04-20		10:00:00-11:30:00		\$12.0
15		exhibition02		2026-04-20		10:00:00-11:00:00		\$7.5
16		Ancient civilizations		2026-04-20		12:00:00-13:30:00		\$12.0
17		Renaissance masters		2026-04-20		12:00:00-13:30:00		\$12.0
18		exhibition02		2026-04-20		12:00:00-13:00:00		\$7.5
19		Ancient civilizations		2026-04-20		14:00:00-15:30:00		\$12.0
20		Renaissance masters		2026-04-20		14:00:00-15:30:00		\$12.0
21		exhibition02		2026-04-20		14:00:00-15:00:00		\$7.5

In order to complete your reservation, we need some information.

Select a tour by typing the option number.

5

Insert your email address:

[zach4000@yahoo.com

Reservation created successfully!

Screenshot of the insert:

[db2 => SELECT * FROM RESERVATION;

EMAIL

TOUR_ID

DATE

START_TIME

BOOKING_DATE

cnani0@studiopress.com

1 07/05/2025 10:00:00 05/30/2025

...
zach4000@yahoo.com

5 04/15/2026 10:00:00 03/30/2026

29 record(s) selected.

db2 => □

Option 3: Relocate (or place for the first time) an artifact that is in an exhibition to a new room

```
[Select an option: 3
You have selected to relocate an artifact.
In order to complete the relocation, we need some information.
Insert the artifact ID:
[20
Insert the exhibition ID that the artifact belongs to:
1
Insert the new room:
[Willy wonka room
Artifact relocated successfully!
```

Screenshot proof of insert:

```
[db2 => SELECT * FROM DISPLAYS WHERE ARTIFACT_ID = 20;

ARTIFACT_ID EXHIBITION_ID ROOM
-----
          20             1 Willy wonka room

1 record(s) selected.
```

Option 4: View tour statistics ordered by total revenue and number of reservations

Group 22 Museum Main Menu:

1. Search for an artifact
2. Make a reservation for a tour
3. Relocate an artifact
4. View tour statistics
5. Add a new exhibition
6. Quit

```
[Select an option: 4
```

Tour ID	Number of reservations	Total revenue
4	13	156.00
2	6	120.00
1	4	60.00
3	3	51.00
5	6	45.00

Option 5: Add a new exhibition

```
Select an option: 5
In order to add an exhibition to the museum, we need some information.
Insert the exhibitions name:
Modern Feminism
Insert the exhibitions description:
A walkthrough feminist art of the 21st century
Insert the start date of the exhibition (YYYY-MM-DD):
2026-04-01
Insert the end date of the exhibition (YYYY-MM-DD):
2026-08-01
Exhibition created successfully!
In order to complete the exhibition creation, please add all genres associated with the exhibition.
For your convenience, this is the list of all genres :
Abstract
Armory
Baroque
Clothing
Coins and medals
Contemporary
Decorative art
Egyptian
Gothic
Hellenistic
Impressionism
Japanese Edo-period
Jewels
Mannerism
Neoclassic
Realism
Religious
Renaissance
Rococo
Romanticism
Sumerian
Insert a genre name:
Realism
Genre added successfully!
Insert a genre name or 'DONE':
Neoclassic
Genre added successfully!
Insert a genre name or 'DONE':
Hellenistic
Genre added successfully!
Insert a genre name or 'DONE':
DONE
Exhibition created successfully!
```

Screenshot proof of the insert:

```
db2 => SELECT * FROM EXHIBITIONS WHERE NAME = 'Modern Feminism';
EXHIBITION_ID NAME
              DESCRIPTION
              START_DATE END_DATE
-----
14 Modern Feminism A walkthrough feminist art of the 21st century
04/01/2026 08/01/2026
1 record(s) selected.
```

Indexing

Script executing create index statement:

```
cs421g22@winter2026-comp421:~$ db2 "CREATE INDEX artist_name_index ON Artists(name)"
DB20000I The SQL command completed successfully.
```

Queries that will run faster with this index:

Queries 1 and 3 require sorting the final result in order of artist name. Without the index, the database must perform a full table scan and sort the artists in alphabetical order. With this

new index, the database can simply go through the artist name index in order, as the artist names are already stored in the leaf nodes in alphabetical order.

Furthermore, Query 3, which gets the top 10 most collaborative artists by number of artifacts, joins the Artists table to the ArtistCollabCounts table. Adding this index allows the database to quickly locate the specific name string associated with each artist_id found in the collaboration set.

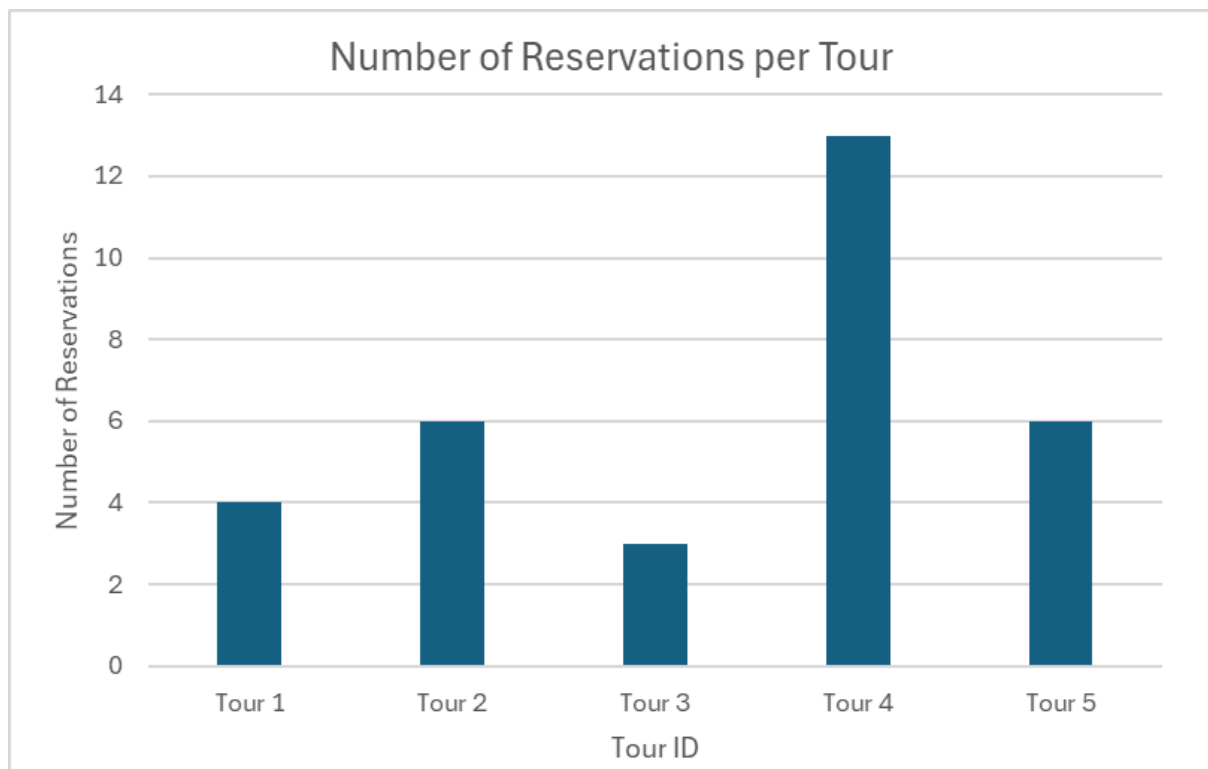
Lastly, if visitors want to search for a specific artist, this index prevents the database from having to scan through the full artist table, significantly reducing the search latency even if the number of artists increases.

Visualization

SQL used to generate data

```
EXPORT TO reservations_per_tour.csv OF DEL MODIFIED BY NOCHARDEL  
  
SELECT 'Tour ' || tour_id, COUNT(*)  
  
FROM Reservation  
  
GROUP BY tour_id;
```

Image of chart



Explanation of what the graph represents

This bar chart shows the number of reservations for each tour offered by the museum. It plays an important role in the business side of the museum, as it allows them to analyze visitor demand and identify the most popular tours. Much like in a real museum, the information provided by this visualization can help with business decisions, such as adding tour dates for the most frequently booked tours to maximize revenue, promote tours with fewer reservations, etc.

Creativity

Description

The trigger enforces reservation limits for scheduled tours and prevents overbooking. Each tour has a maximum capacity of 8 visitors. When a user attempts to make a reservation for a tour, the trigger counts the current number of reservations for that tour, date, and start time. If there are fewer than 8 reservations, the reservation will be made and inserted to the table. If 8 reservations already exist, the trigger raises an error and prevents the new reservation from being made. This ensures that no guided tour is overbooked and that each tour respects its visitor capacity.

Code

```
CREATE TRIGGER prevent_overbooking
BEFORE INSERT ON Reservation
REFERENCING NEW AS NEW
FOR EACH ROW
BEGIN
    DECLARE reservation_count INT;

    SELECT COUNT(*) INTO reservation_count
    FROM Reservation
    WHERE tour_id = NEW.tour_id
    AND date = NEW.date
    AND start_time = NEW.start_time;

    -- max capacity is 8 for all tours
    IF reservation_count >= 8 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Tour is fully booked';
    END IF;
END @
```

Execute trigger

```
cs421g22@winter2026-comp421:~$ db2 "SELECT tour_id, date, start_time, COUNT(*) AS num_of_reservations FROM Reservation WHERE tour_id = 4 AND date = '04/15/2026' AND start_time = '12:00:00' GROUP BY tour_id, date, start_time"

TOUR_ID    DATE          START_TIME    NUM_OF_RESERVATIONS
-----
4 04/15/2026 12:00:00 5

1 record(s) selected.

cs421g22@winter2026-comp421:~$ db2 "INSERT INTO Reservation VALUES ('enani0@studiopress.com', 4, '04/15/2026', '12:00:00', '01/10/2026'), ('nathan222@shinystat.com', 4, '04/15/2026', '12:00:00', '02/02/2026'), ('slaviste9@nhs.uk', 4, '04/15/2026', '12:00:00', '03/03/2026');"
DB20000I The SQL command completed successfully.
```

The trigger executes before a reservation is inserted. Here, the insert statements executed successfully as the max capacity for the tour has yet to be reached.

Effect of trigger

```

cs421g22@winter2026-comp421:~$ db2 "SELECT tour_id, date, start_time, COUNT(*) AS num_of_reservations FROM Reservation WHERE tour_id = 4 AND date = '04/15/2026' AND start_time = '12:00:00' GROUP BY tour_id, date, start_time"

TOUR_ID    DATE        START_TIME  NUM_OF_RESERVATIONS
-----
4 04/15/2026 12:00:00      8

1 record(s) selected.

cs421g22@winter2026-comp421:~$ db2 "INSERT INTO Reservation VALUES ('sferrullid@sogou.com', 4, '04/15/2026', '12:00:00', '02/03/2026')"
DB21034E  The command was processed as an SQL statement because it was not a
valid Command Line Processor command. During SQL processing it returned:
SQL0438N  Application raised error or warning with diagnostic text: "Tour is
fully booked".  SQLSTATE=45000
cs421g22@winter2026-comp421:~$ db2 "SELECT tour_id, date, start_time, COUNT(*) AS num_of_reservations FROM Reservation WHERE tour_id = 4 AND date = '04/15/2026' AND start_time = '12:00:00' GROUP BY tour_id, date, start_time"

TOUR_ID    DATE        START_TIME  NUM_OF_RESERVATIONS
-----
4 04/15/2026 12:00:00      8

1 record(s) selected.

```

The maximum capacity of the tour has been reached (8 reservations). The trigger prevents further reservations to be made.

Contributions

Our team held two meetings, during which we divided the tasks evenly and collectively reviewed the project deliverable and report prior to submission. Here are each member's contributions:

- Gonzalo: Java and JDBC connection setup. Menu, queries 1 and 2.
- Annabelle: Stored procedure and indexes
- Jessie: Data visualization and trigger
- Clementine: Menu options 3, 4, and 5

All team members contributed to proof reading.